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Green exercise versus indoor urban exercise: Related frontal brain thickness and cognitive performance

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ABSTRACT

Background and aims: This study compared, for the first time, cognitive function, and the cortical thickness of frontal areas in healthy adults who exercise regularly in outdoor green spaces ("green exercise group," GEG; n = 22, mean age = 39.91 years, females = 11) with those who exercise in indoor urban settings ("urban exercise group," UEG; n = 25, mean age = 40 years, females = 11). We further explored the associations between frontal brain thickness and cognitive measures in each group separately.
Method: Forty-seven healthy adults participated. All underwent a neuropsychological evaluation for short-term memory, working memory, and attention. High-resolution T1-weighted 3D magnetic resonance imaging (MRI) scans were acquired for each participant. The thickness of frontal brain regions was extracted and exported to JASP software for statistical analyses.
Results: The GEG performed better on working memory and short-term memory tasks compared to the UEG. Additionally, the GEG exhibited increased thickness in the right anterior cingulate cortex (rACC) and the right superior frontal sulcus (rSFS). These areas were associated with short-term memory within the GEG. Fisher's test and Zou's intervals revealed that only the correlation between the rSFS and short-term memory was significantly different between the UEG and GEG.
Conclusion: These findings suggest that exercising in outdoor natural areas might be related to better cognitive function and greater brain thickness in frontal areas. This study will contribute to the growing body of evidence linking brain, cognition and exercising in natural areas. Further research is needed to confirm these findings and explore the underlying mechanisms.

1. Introduction

Recently, several initiatives worldwide, such as "Healthy Parks, Healthy People", are demonstrating the importance of interacting with green spaces for maintaining healthy cognitive function. According to the Attention Restoration Theory (ART; Kaplan, 1995), and supported by previous studies (Berman et al., 2012; Berto, 2005; Ohly et al., 2016; Stevenson et al., 2018), even brief exposure to natural environments can improve cognitive performance. This includes benefits such as a better working memory (Dadvand et al., 2015; Ohly et al., 2016; Stevenson et al., 2018), short-term memory (Berman et al., 2012; Ohly et al., 2016), attentional control (Dadvand et al., 2018) sustained attention (van den Berg et al., 2003), and even structural and functional brain changes in prefrontal and premotor areas (Beyer et al., 2024; Dadvand et al., 2018). Interestingly, even simply observing natural environments, without direct interaction, has been shown to benefit directed sustained attention (van den Berg et al., 2003), working memory (van Oordt et al., 2023) and brain function (Kim et al., 2010). However, the positive impact of nature on cognition varies with the type of interaction. Some studies suggest higher benefits when exposure to green open spaces is combined with physical activity, generating what's called green exercise (GE). Recent research proposes that combining PA with nature exposure might have synergistic effects, further enhancing cognitive function (Cerin et al., 2021; Lachowycz & Jones, 2013). Supporting this notion, several studies have identified differences in various psychological outcomes between exercising in urban environments and green outdoor

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exercise (GE), suggesting that engaging in GE may yield greater benefits (Bodin & Hartig, 2003; Lahart et al., 2019). Specifically, research indicates that exercising in natural environments can positively impact attention, working memory (Han, 2017), inhibitory control (Bailey et al., 2018) and cognitive reserve (D'Aurizio et al., 2023). However, some studies report mixed results for working memory, finding no correlation or even a negative association with GE (Jiménez et al., 2022; McKendrick et al., 2017). Despite these mixed findings, the little existing evidence on GE and cognition suggests a potential link between exposure to green spaces and improved cognitive function (Dadvand et al., 2015; Jiménez et al., 2022). Nevertheless, not all green spaces elicit the same effects. Notably, outdoor natural environments, such as forests, mountains or valleys, may yield more pronounced positive psychological effects than both indoor settings (Kerr et al., 2006, as cited in Bowler et al., 2010; Thompson Coon et al., 2011) and urban greenspaces (Li et al., 2022). This discrepancy between the effects of natural and urban greenspaces can be attributed to the fact that urban greenspaces, often characterized by semi-natural environments, are exposed to various detrimental health factors, including noise pollution, overcrowding, and air pollution, which may mitigate the beneficial impacts of green exposure (Li et al., 2022).

In relation to brain benefits of physical activity (PA), a variety of studies have demonstrated a link between PA, regardless of setting, and improvements in cognitive performance (Chang et al., 2012; Pesce et al., 2021) and cognition and brain function and structure (Erickson et al., 2014; Tseng et al., 2013; Voelcker-Rehage & Niemann, 2013). Notably, research has implicated the frontal brain regions in both supporting higher-level cognitive function (Erickson et al., 2014; Ruscheweyh et al., 2011) and mediating the beneficial effects of PA on such mental functions (Bento-Torres et al., 2019; Weinstein et al., 2012). Indeed, Ruscheweyh et al. (2011) and Weinstein et al. (2012), suggest that increased physical activity (PA) may lead to improved cognitive function by influencing brain structure, particularly the prefrontal cortex (PFC). This hypothesis is supported by research finding associations between higher PA levels and greater activation of both frontal and parietal regions during cognitive tasks (Colcombe et al., 2004; Prakash et al., 2011). Studies like Colcombe et al. (2006) specifically linked PA to increased volume in the anterior cingulate cortex (ACC) and inferior and middle frontal gyrus (IFG; MFG), findings echoed in subsequent investigations (Bashir et al., 2021; Bento-Torres et al., 2019; Herting et al., 2016; Ruscheweyh et al., 2011). The dorsolateral prefrontal cortex (dlPFC) also appears to be affected by PA, with research consistently demonstrating a positive correlation between PA and DLPFC volume (Northey et al., 2020; Weinstein et al., 2012).

Interestingly, reduced gray matter volume in these PFC regions is associated with various cognitive challenges throughout life, including lower academic performance in young people (Urrila et al., 2017) and cognitive decline in older adults (Bell-McGinty et al., 2005). Conversely, larger PFC volume and thickness are linked to enhanced executive function (Yuan & Raz, 2014). While these findings provide strong evidence for PA's positive impact on cognition, it's important to acknowledge that, as far as we know, no study to date has investigated the relationship between green exercise, cognition and the cortical thickness of the prefrontal cortex. Therefore, no previous results can be shown in this regard. This study will contribute to the growing body of evidence on this topic.

This study aims to explore the potential associations of green exercise (GE) with cognitive function and brain structure compared to urban exercise. Our first objective was to compare a group of healthy adults who regularly practice GE to another group exercising in urban indoor settings, focusing on differences in cognitive performance (short-term memory, working memory, and sustained attention and attentional control). Besides, we explored the differences in the frontal lobe cortical thickness. Finally, we also conducted a complementary analysis to investigate potential associations of the significant between-groups differences in frontal cortical thickness and cognitive measures within each group.

We hypothesized that participants who exercised in natural environments would exhibit superior cognitive performance in attention control (Dadvand et al., 2018), sustained attention (Berto, 2005), working memory (Dadvand et al., 2015; Ohly et al., 2016; Stevenson et al., 2018), and short-term memory (Ohly et al., 2016) compared to those who exercised in indoor urban areas. Besides, given that previous studies have demonstrated differences in frontal brain structure associated with exposure to nature (Beyer et al., 2024; Dadvand et al., 2018), we expect to find between-group differences in the frontal brain cortical thickness. However, a lack of studies in cortical thickness measures in the context of green exercise preclude concrete hypotheses about the specific directions or regions that may exhibit differences or about the associations of specific brain structure with cognitive measures.

2. Methods

2.1. Participants

We recruited a voluntary response sample of 49 adult participants in Granada (south of Spain). This sample size is comparable to those used in prior research investigating the associations between brain metrics and physical activity (Bento-Torres et al., 2019; Colcombe et al., 2006; Herting et al., 2016) and between brain metrics and nature exposure (Bratman et al., 2015; Sudimac et al., 2022). Two participants were excluded from subsequent analysis because they did not complete the MRI scanning. The remaining participants (47) were divided into two groups regarding their preferred areas for exercising: those who exercised in outdoor natural areas (green exercise group; GEG; n = 22; mean age = 39.91; females = 11) and those who exercised in urban indoor settings (urban exercise group; UEG; n = 25; mean age = 40; females = 11). Inclusion criteria included (i) ages between 18 and 65, (ii) exercising at least three times a week, and (iii) a minimum of 5 years of experience practicing sport in either natural outdoor or urban indoor setting.

All participants were asked about the sports activities they engaged in and the specific locations where they performed these activities. To be included in the green exercise group, participants had to engage in outdoor sports activities in wild nature. This included mountains, valleys, or forests, among other locations. Urban greenspaces, such as city parks, were not considered natural areas due to the previously demonstrated differences between wild natural spaces and urban greenspaces (Li et al., 2022). Participants in the urban indoor exercise group were required to engage in activities conducted in facilities situated within city limits. These facilities had to be located in such a way that any views from windows, if present, did not encompass green spaces but rather urban scenes such as streets, avenues, buildings or industrial zones. Examples of suitable facilities include indoor gyms and sports centers. In the rare instances that individuals in UEG utilized an outdoor urban facility (e.g., city hiking), these activities were conducted in commercial areas, public squares, or along urban streets and their itinerary was carefully asked to be sure they consistently avoided green parks. Specifically, two participants engaged in outdoor urban exercise in UEG, specifically city hiking. However, they always confined their activities within city limits to maintain an exclusively urban context (see Table A.1 in the Supplementary Material). Moreover, this was not their primary or most frequent form of exercise. The use of urban outdoor facilities by the remaining participants was sporadic. The physical activities performed by the participants can be consulted in Table A.1. (Appendix A, Supplementary Material). If any participants of GEG or UEG practiced exercise in any urban or green setting, respectively, the frequency were to be equal or less than once a month. In both cases, water sports were excluded to avoid confusion with the natural environment (Grassini et al., 2019). Exclusion criteria were: (i) having any psychological disorder or any medical or nervous system disease diagnosed, (ii) taking any medication regularly, (iii) having suffered an

epileptic attack, (iv) taking any drugs, and drinking high amounts of alcohol regularly, and (v) having any contraindication to undergo the MRI session (i.e., metal prosthesis or claustrophobia). Participants were informed about the study's objectives and signed an informed consent. After that, all the participants performed a Magnetic Resonance Imaging (MRI scan) and completed a neuropsychological assessment with several tasks. It is worth noting that none of the participants (regardless of group) lived far from the natural area. Only 4 participants (in the UEG group) lived more than 1 km away from a natural area. This information is detailed in Table 1.

The data from each participant was collected in a same-day session of approximately 1 h and 20 min in total duration. All participants first underwent MRI scanning, followed by neuropsychological testing to assess cognitive measures (short-term memory, working memory and attention). Participants did not receive any compensation for their participation in the study.

The Human Research Ethics Committee of the University of Granada approved the study (621/CEIH/2018), and all the participants were informed about the aim of the study and signed an informed consent.

2.2. Measures

2.2.1. Sociodemographic and sport characteristics

A questionnaire about sociodemographic and characteristics of the physical activity practiced was completed in the first session. This questionnaire assessed:

Table 1
Demographics and clinical characteristics of the groups.

	GEG (N = 22)	UEG (N = 25)		
	Mean (SD)/ Freq (%)	Mean (SD)/ Freq (%)	Test statistics [a]	*p* value
Age	39.91 (6.35)	40 (6.19)	0.000	1.000
Sex (females)	11 (50%)	11 (44%)	0.169	0.681
DNNA (meters)				
0–500 m	19 (86.4%)	15 (60%)	5.803	0.055
500 m-1 km	3 (13.6%)	5 (20%)		
1 km–3 km	0	5 (20%)		
Dwelling place (years)	9.273 (2.271)	12.780 (2.692)	−0.981	0.332
Sport (level)				
High	12 (54.5%)	7 (28%)	5.719	0.057
Medium	7 (31.8%)	7 (28%)		
Low	3 (13.6%)	11 (44%)		
Muscular mass (kg)	52.21 (10.36)	53.81 (11.91)	−0.490	0.626
Fat mass (kg)	21.62 (6.91)	23.19 (7.7)	−0.733	0.467
Education (level)				
Primary	0	2 (8%)	1.842	0.398
High school	3 (13.6%)	3 (12%)		
University	19 (86.4%)	20 (80%)		
Tobacco smoking				
Daily	1 (4.5%)	2 (8%)	0.238	0.888
Occasionally	1 (4.5%)	1 (4%)		
Never	20 (90.9%)	22 (88%)		
Alcohol drinking				
Daily	2 (9.1%)	0	3.542	0.315
Weekly	4 (18.2%)	3 (12%)		
Occasionally	14 (63.6%)	17 (68%)		
Never	2 (9.1%)	5 (20%)		
Short-term memory	8.0 (1.38)	5.60 (1.29)	6.158	<0.001
Working memory	7.64 (1.73)	5.52 (1.85)	4.029	<0.001
Attention control d2	70.41 (22.70)	72.44 (22.67)	−0.306	0.761
Concentration d2	62.73 (27.01)	69.36 (24.69)	−0.879	0.384

Note: *Means and standard deviations are shown for quantitative variables, while frequencies and percentages are used for ordinal and nominal variables. Abbreviations: GEG, green exercise group; UEG, urban exercise group; DNNA: distance to nearest natural area. Dwelling place (years) refers to the number of years living in the same place.

- Sociodemographic variables, including age, sex, education level, smoking status, and alcohol consumption. The participants' addresses were also collected to calculate the distance between their residences and the nearest green area.
- Anthropometric variables such as body muscular and fat mass.
- Sport type and frequency. Participants were asked about their preferred sports/activities and their preferred setting (urban or green) to exercise. This information is shown in Table A.1. in Appendix A (Supplementary Material). Frequency of physical activity was also asked. We ensured all participants engaged in physical activity at least three times per week.
- Sport level. This measure includes an index of the intensity of the PA. This variable was divided into three groups: High sport level includes participants with a federation or membership card who normally participate on competitions. Medium level includes participants with a federation card that do not normally participate in competitions. Low level includes participants with a weekly sport practice but without a federation or membership card.

2.2.2. Imaging data acquisition

An MRI scan was carried out for each participant in a 3T Tim Trio Siemens with a 32-channel phased array head coil sited at the Mind, Brain and Behavior Research Center, University of Granada. We acquired a high-resolution T1-weighted 3D magnetization-prepared rapid gradient-echo (MPRAGE) sequence for each participant with the following parameters: Repetition Time (TR) = 2.3 ms, Echo Time (TE) = 3.1 ms, flip angle = 9°, Field of View (FOV) = 256 × 256 mm, matrix size = 320 × 320, number of slices = 208, resulting an isotropic resolution of 0.8 × 0.8 × 0.8 mm. The total acquisition time for the T1 sequence was 6 min and 35 s. In this study, only cortical thickness metrics were used to follow the study's objectives.

2.2.3. Cognitive tasks

Digit Ordering Test (Werheid et al., 2002). This test assesses short-term memory, which refers to the ability to hold a small amount of information for a brief period (Gathercole, 1999). Participants listen to a sequence of randomly generated digits and are asked to immediately repeat them back in the same order presented by the examiner. The digits are presented at a rate of one per second, mimicking natural speech rhythm. If participants respond correctly, they progress to the next trial with a longer sequence. The test stops after two incorrect responses at the same sequence length. The maximum achievable score is 15. A version of this task has been used in some studies exploring associations between nature exposure and cognition (Perkins et al., 2011; Stark, 2003; cited in Ohly et al., 2016).

Backwards Digit Span Task (Wechsler, 1999). The "backward digit span" task was used to assess working memory, which refers to the ability to hold onto information while performing other mental tasks (Baddeley, 2010). In this task, the examiner read a sequence of numbers aloud, one digit per second. On each trial, participants repeated the numbers in the reverse order they were presented. The task ended after two consecutive errors at the same span length. The total score reflected the number of correctly recalled sequences. The maximum achievable score was 14. This task has been used in similar studies investigating the association between cognition and nature (Han, 2017; Jonides et al., 2008).

The d2 Test of Attention (Brickenkamp, 2002). The d2 Test of Attention measures sustained attention, selective attention, attention control and processing speed. It's a widely used neuropsychological tool for assessing attentional functioning in both children and adults and represents a convenient tool to measure different domains of attention in a short period of time. The test consists of a sheet filled with rows and columns of interspersed d's and p's (or other letters or symbols). Participants are instructed to mark all the d's (or specific targets) within a time limit, typically 4 min. Performance is evaluated based on several factors. For this study, we used the total score of attentional control

(TOT) and the concentration index (CON). This task has been frequently used in similar studies investigating the association between cognition and nature (Lindemann-Matthies, Benkowitz, & Hellinger, 2021; Pisalski et al., 2020).

These cognitive tasks were counterbalanced to ensure that any potential imbalance across the different tasks was minimized.

2.3. Analyses

2.3.1. Preprocessing and parcellation of imaging data

All images were manually inspected for major artefacts and realigned to the AC-PC line. Image processing was conducted using the automated processing pipeline in FreeSurfer software (version 6.0) on the Alhambra cluster of the University of Granada, Spain. Preprocessing steps included intensity normalization, registration to Talairach space, skull-stripping, white matter segmentation, tessellation of the white matter boundary, and automatic correction of topological defects. Cortical thickness was then estimated using a procedure described by Fischl and Dale (2000) that calculates the distance between the reconstructed gray-white matter and the gray-pial boundary surfaces. The distance provides the thickness of the cortical gray matter with submillimeter accuracy at any point (Fischl & Dale, 2000).

The cerebral cortex was parcellated into 148 regions of interest (ROIs) across both hemispheres based on gyral and sulcal structures from the Destrieux atlas (Destrieux et al., 2010; Fischl & Dale, 2000; Fischl et al., 1999). The ROIs are then grouped into five lobes. We focused on the prefrontal regions due to their consistent association with both high-order cognitive functions and the brain benefits of physical exercise (Bashir et al., 2021; Bento-Torres et al., 2019; Erickson et al., 2014; Flöel et al., 2010; Herting et al., 2016; Ruscheweyh et al., 2011, among others). Seventeen ROIs were selected for this study. The numbers in brackets correspond to the numbering of frontal brain areas in Destrieux et al. (2010): frontomarginal gyrus and sulcus [1], transverse frontopolar gyrus and sulcus [5], anterior cingulate gyrus and sulcus [6], middle frontal gyrus [15], middle frontal sulcus [53], superior frontal gyrus [16], superior frontal sulcus [54], inferior frontal gyrus (divided into triangular [14], opercular [12], and orbital parts [13]), and inferior frontal sulcus [52], orbital gyrus [24], rectus gyrus [31], lateral orbital sulcus [62], medial olfactory sulcus [63], the orbital H-shaped sulcus [64] and the suborbital sulcus [70]. These 17 ROIs were extracted for each hemisphere. We excluded the subcentral and precentral areas from consideration due to its primary involvement in the motor system. Cortical thickness within these prefrontal regions, based on the Destrieux atlas, was extracted, and exported to an Excel spreadsheet.

2.3.2. Subsequent analyses

The analyses were conducted using JASP software (version 16.3). To explore whether the cognitive variables and the frontal cortical thickness differed between groups, two independent sample t-tests were carried out. Additionally, we performed complementary ANCOVA models to examine the between-group differences and check if our covariates (age, education, sport level, and intracranial volume) had a statistical influence over the differences in the cortical thickness of different frontal regions. Finally, separate partial correlations (within each group) were performed using the same software to examine the associations between regions showing between-group differences in frontal cortical thickness and cognitive variables. These analyses were conducted separately for each participant group to examine intra-group associations. To account for potential confounders, we included sex, age, sport level, and education level in the correlation models. Finally, Zou's intervals and Fisher's tests were employed through the cocor package (Diedenhofen & Musch, 2015) to assess two-tailed correlations for independent samples and test for differences in correlation coefficients between groups (GEG versus UEG). Results were considered statistically significant at $p < 0.05$, or, in the case of confidence intervals, when the interval did not include zero.

3. Results

3.1. Descriptive analyses

Groups did not differ significantly in their education level, sport level, muscular and fat mass or in their smoking or drinking frequency. This information is detailed in Table 1.

3.2. Between-group differences in cognitive measures and frontal thickness

Groups differed significantly in their memory performance, namely, working memory ($t = 4.046$, $p < 0.001$, 95%CI [1.124, 3.078]) and short-term memory ($t = 6.131$, $p < 0.001$, 95%CI [1.599, 3.085]), with the GEG showing higher scores (Table 1). There were no differences in any index of attention (Table 1). The GEG showed greater thickness in both the right anterior cingulate (right ACC; $t = 2.181$, $p = 0.034$, 95%CI [0.01, 0.165]) and the right superior frontal sulcus ($t = 2.151$, $p = 0.037$, 95%CI [0.01, 0.147]). All results, including non-significant ones, are presented in Tables B.1 and B.2 of Appendix B (Supplementary Material).

3.3. Complementary analysis of variance for the between-groups differences in cortical thickness

Regarding between-group differences in the right ACC thickness, age did not significantly predict the between-groups differences ($F = 2.64$, $p = 0.111$), and the differences remained non-significant ($p = 0.039$). Additionally, sex did not significantly predict the between-group differences ($F = <0.001$, $p = 0.977$) and the differences remained significant ($p = 0.037$). Similarly, sport level did not significantly predict the between-group differences either ($F = 0.47$, $p = 0.498$) and they remained significant ($p = 0.028$). Furthermore, intracranial volume did not significantly predict the between-group differences ($F = 1.28$, $p = 0.265$). The differences remained to be significant ($p = 0.034$). Finally, education was the only variable significantly associated with between-group differences in ACC ($F = 4.283$, $p = 0.044$). However, after controlling for education, between-group differences in ACC cortical thickness remained significant ($p = 0.015$), indicating that these differences are not solely attributable to educational level.

Regarding the cortical thickness of the right superior frontal sulcus, the results indicated that none of these variables were associated with the between-group differences. Specifically, age did not significantly predict the between-groups differences ($F = 0.29$, $p = 0.595$), and the differences remained non-significant ($p = 0.039$). Sex did not significantly predict the between-group differences either ($F = 1.08$, $p = 0.305$) and the differences remained significant ($p = 0.032$). Similarly, sport level did not significantly predict between-group differences ($F = 0.02$, $p = 0.887$) and they remained significant ($p = 0.047$). Moreover, intracranial volume did not significantly predict the between-group differences ($F = 0.045$, $p = 0.833$). The differences remained to be significant ($p = 0.039$). Finally, Education did not significantly predict the between-group differences ($F = 0.053$, $p = 0.819$). The differences remained to be significant ($p = 0.044$).

3.4. Partial correlations between cognitive measures and frontal thickness in each group separately controlled by age, sex, education and sport level

Those who exercised in natural outdoor areas (GEG, $N = 22$) showed a direct positive association between short-term memory and the thickness of the right ACC [$r(16) = 0.498$, $p = 0.035$, 95%CI (−0.075, 0.861)] and the right superior frontal sulcus thickness [$r(16) = 0.541$, $p = 0.020$, 95%CI [0.007, 0.829]. Although, p-value did confirm a significant association, confidence intervals showed that association

between right ACC and short-term memory did not surpass the significance threshold. No significant association was found in UEG (N = 25). These results are depicted in Table 2 for GEG, and Table 3 for UEG.

3.5. Between-group correlation differences (Fisher's test and Zou's intervals)

Fisher's test and Zou's intervals confirmed that the correlation between the short-term memory – right superior frontal sulcus is significantly different between the groups (p = 0.005, 95%CI = 0.232, 1.210), while the short-term memory – right ACC correlation was not significantly different between GEG and UEG (p = 0.299, 95%CI = −0.7681, 0.250).

4. Discussion

The main purpose of the study was to explore the differences in the frontal lobe cortical thickness and cognitive measures between individuals who exercise while exposed to green outdoor nature (green exercise group; GEG) and individuals who exercise in indoor urban settings (urban exercise group; UEG). Our findings showed that individuals in the GE group showed better working memory (WM) and short-term memory. These results are in line with the studies of Berman et al. (2012) and Stevenson et al. (2018) who found better scores in working memory tasks in those individuals who frequently interact with natural environments. Lega et al. (2021) found that surrounding greenness is related to better overall memory performance and a specific better short-term memory score. Regarding WM, research has yielded mixed findings. Our study identified between-group differences, with GEG demonstrating superior WM performance. This aligns with previous research by Dadvand et al. (2015) and Stevenson et al. (2018), but contradicts studies like McKendrick et al. (2017), which reported negative associations in certain WM dimensions. Additionally, other research (Jiménez et al., 2022) has found no association between these two variables. A potential explanation for the discrepancies between our study and those with negative results could be the inclusion of physical activity in our research. The abovementioned studies resulted from the passive interaction of the individuals with greenness and/or natural environments, while in our study, individuals took an active role in the interaction (practicing sports). This active role added to the interaction with greenness might result in greater use of different cognitive abilities related to exercising, such as working memory (Lambourne, 2006; Padilla et al., 2014). On the other hand, we did not observe any significant differences in attention measures. This was unexpected and contradicts previous studies (Dadvand et al., 2018; McDonnell & Strayer, 2024; among others). Although previous studies had reported superior performance in selective attention (Laumann et al., 2003), reaction time or detection sensitivity (Berto, 2005) among urban groups compared to nature groups, a recent review found no association between green exercise and attention (Lahart et al., 2019). A possible explanation for these inconsistent results may be found in Tilley et al. (2017). Their research demonstrated that walking in urban and busy environments increased attention recruitment due to the stressful and resource-demanding nature of these spaces. Likewise, Bailey et al. (2018) found that natural outdoor settings can induce a more meditative state, as they are less attention demanding. Grassini et al. (2019) supported these results, finding that pictures of natural spaces demanded less attentional and cognitive load. Future studies should investigate both external attention and attention awareness in individuals exercising in natural environments to clarify this potential difference.

Table 2
Pearson's partial correlations in the green exercise group (GEG).

Variable	1	2	3	4	5
1. R_sFS	–				
2. R_ACC	**0.630****				
3. stM	**0.541***	**0.498***			
4. WM	0.405	0.175	**0.752****		
5. TOT	−0.227	0.231	0.119	0.008	
6. CON	−0.126	0.324	0.119	0.019	**0.932****

Note: *Partial correlations conditioned on sex, age, sport level and education level. Abbreviations*: R_sFS, right superior frontal sulcus; R_ACC, right anterior cingulate cortex; *sTM, short term memory; WM, working memory; TOT, total attention score; CON, concentration score. * for p ≤ 0.05; ** for p ≤ 0.01.*

Table 3
Pearson's partial correlations in the green exercise group (UEG).

Variable	1	2	3	4	5
1. R_sFS	–				
2. R_ACC	−0.061				
3. stM	−0.255	0.218			
4. WM	0.051	−0.037	0.326		
5. TOT	−0.318	0.111	0.112	<0.001	
6. CON	−0332	0.015	0.029	0.156	**0.720****

Note: *Partial correlations conditioned on sex, age, sport level and education level. Abbreviations*: R_sFS, right superior frontal sulcus; R_ACC, right anterior cingulate cortex; *sTM, short term memory; WM, working memory; TOT, total attention score; CON, concentration score. * for p ≤ 0.05; ** for p ≤ 0.01.*

Additionally, GEG exhibited greater cortical thickness in the right anterior cingulate (ACC) and right superior frontal sulcus compared to UEG. We also found a direct positive association between short-term memory and the thickness of the right ACC and the right superior frontal sulcus only in GEG. However, we did not find any such association in UEG. These results align with several studies on PA, brain, cognition and nature exposure.

Bento-Torres et al. (2019) demonstrated an association between increased right ACC activation during executive function tasks and cardiorespiratory fitness. On the other hand, Miró-Padilla et al. (2023) found a positive correlation between right ACC volume and self-reported physical activity. Interestingly, they also discovered a negative correlation between ACC volume, reward sensitivity, and sedentary time. Given the ACC's role in decision-making and selecting advantageous options for long-term benefits (Shenhav et al., 2013), Miró-Padilla et al.'s (2023) results suggest that right ACC structure might influence the choice between sedentary behavior and physical activity. However, our results revealed significantly higher right ACC thickness only in individuals engaging in green exercise, and not in those who exercised indoors in urban areas. This suggests that the right ACC might be sensitive to or influenced by the environmental characteristics of exercise. This aligns with Kim et al.'s (2010) finding that the right anterior cingulate cortex is more activated following exposure to natural scenic images compared to urban images. Additionally, studies have linked urban upbringing to reduced cortical thickness in various ACC subregions (Haddad et al., 2015), further supporting the hypothesis that environmental factors can influence ACC structure.

Moreover, some studies support the idea that the superior frontal sulcus is also sensitive to environmental characteristics. Besteher et al. (2017) discovered an inverse association between the cortical thickness of the superior frontal sulcus and an urbanicity index. Other studies have reported similar findings. Dadvand et al. (2018) observed a positive association between lifelong exposure to greenness and larger gray matter in the right superior and inferior prefrontal cortex in schoolchildren and Kim et al. (2010) found a higher activation of the right superior prefrontal cortex after acute exposure to images of nature, compared to urban images. In line with these findings, a recent review (Beyer et al., 2024) demonstrated that long-term urban exposure negatively impacts the dorsolateral and medial prefrontal cortices, which include the ACC and the superior frontal sulcus. This suggests a link between environmental exposure, both long-term and short-term, and prefrontal structure. Overall, our results suggest that the right superior frontal sulcus and the right anterior cingulate cortex may be particularly responsive to prolonged exposure to natural stimuli and physical

exercise.

A notable finding was that both of our results were lateralized to the right hemisphere. These results could be interpreted in terms of the right hemisphere's involvement in physical activity and cognition.

Firstly, our results align with previous studies associating the right part of the ACC with physical activity (Bento-Torres et al., 2019; Miró-Padilla et al., 2023). Indeed, Bento-Torres et al. (2019) showed an association between the volume of the right ACC, greater physical activity and better executive function. This aligns with a previous study by Spagna et al. (2020), which demonstrated the right hemisphere's superiority over the left in tasks requiring orienting, disengaging, conflict resolution and executive control of attention. Further previous studies also demonstrated the right hemisphere's dominance in tasks demanding spatial attention and target detection (Shulman et al., 2010) and inhibitory control (Garavan et al., 1999). These results highlight the importance of the right hemisphere in cognitive processes.

Specifically addressing our resulting brain areas, the right superior frontal sulcus, located within the dorsolateral prefrontal cortex (dlPFC; Drudik & Petrides, 2024), and the ACC are particularly relevant for executive control (Spagna et al., 2020), memory (Henson et al., 1999), attention shifting (Kondo et al., 2004) and problem-solving (Yi et al., 2022). Indeed, Robertson (2014) proposed a model suggesting that cognitive processes such as sustained attention, alerting, orienting, error awareness, and working memory, along with their neural correlates in the right fronto-parietal network, are key mediators of cognitive reserve in adults.

Interestingly, green exercise has also been linked to enhance cognitive reserve in older adults (D'Aurizio et al., 2023) and has been demonstrated to improve attention, working memory (Han, 2017), and inhibitory control (Bailey et al., 2018), among other cognitive processes. One possible mechanism underlying these cognitive benefits is that frequent green exercise may preferentially stimulate the right frontal areas, which are more strongly linked to cognitive functions than the left frontal area (Shulman et al., 2010; Spagna et al., 2020). This stimulation could potentially lead to enhanced cognitive performance (Beyer et al., 2024).

Considering these findings, it is not surprising that our study revealed an association between green exercise, greater right ACC and right superior frontal sulcus thickness, and a better short-term memory performance. These findings are partially consistent with those of Dadvand et al. (2018), who observed associations between green space exposure, greater right ACC volume, and memory and attentional measures. However, unlike Dadvand et al. (2018), our study focused on cortical thickness rather than volume. Additionally, while we identified associations with cognitive measures, particularly short-term memory, we did not find connections with working memory, as Dadvand et al. (2018) did. These differences may arise from the high overlap between short-term memory and working memory abilities. Nonetheless, further studies should investigate the differences between these dimensions and their associations with greenness and PA. Due to the exploratory nature of these results, additional studies are required to provide further scientific evidence.

Besides, despite the general established relationship between ACC and attention control (Kondo et al., 2004; Luks et al., 2002, among others), we did not find any association between a thicker right ACC and attentional control or concentration. These discrepancies could be attributed to variations in dimensions and indexes assessed between studies, but also to a more theoretical issue. The ACC is a brain area implicated not only in the Salience Attention Network (SAN), but also in the Default Mode Network (DMN). According to Grassini et al. (2019) or Bailey et al. (2018), exposure to nature leads to relaxation, meditative states and internal attention. Therefore, we can speculate that the ACC may be more connected to the other areas forming the DMN (posterior cingulate area, precuneus, among others) to lead to a more relaxed state, rather than to the areas forming the SAN (mainly, the anterior insula) in the context of green exposure (GEG) in our sample. Future studies should investigate the activation and connectivity of DMN and SAN in individuals exposed to nature compared to a control group.

5. Conclusions

Our study has several limitations. In the first place, the cross-sectional nature of our data did not allow us to explore cause-effect relationships of our variables. Including another group of individuals exclusively exercising in outdoor urban facilities would have facilitated stronger conclusions. Secondly, evaluation of exercising duration has not been carried out in this study. Evaluating the duration of the physical activity or the expertise performed in the activity in a more quantitative way might be more useful to better understand the extent to which the type or duration of exercise in green nature might have differential benefits on brain and cognition. Another flaw of this study is that our limited sample size did not allow us to divide the participants into groups according to the characteristics of their preferred PA (aerobic vs. anaerobic) or the intensity of these sports (low vs. high). This prevented us from considering the influence that different activities or their specific intensity might have on the differences observed in the results. It might be interesting to explore the impact of each activity differently on brain and cognitive function. Additional research could also focus on comparing groups engaged in a more comparable form of physical activity. might also aim to compare groups with a more homogeneous type of physical activity. Besides that, the voluntary response sampling performed to recruit the sample might induce a self-selection bias that may impact the results of the study. We acknowledge that our study did not capture the motivations behind participants' preferred exercise setting (green or urban). Future research should investigate the individual and socioeconomic factors influencing this preference. Another potential limitation of our study is the restriction of our analyses to solely frontal brain regions. The exclusion of subcortical areas, such as the hippocampus, which is highly responsive to physical activity (Firth et al., 2018), may have limited our ability to identify broader neural effects. Finally, we did not preregister the present study. Our study, however, has many strengths. It examined the association between cognition, brain structure, PA, and long exposure to green areas from a new and comprehensive perspective, using different methodologies such as neuropsychological assessment, physical evaluations, and neuroimaging techniques. Finally, to our knowledge, this is the first study to find that practicing sport in natural areas, compared to exercising in urban indoor settings might be related to differences in cognitive functioning and in their potential neural substrate: greater thickness in frontal brain areas related to cognition.

The main conclusion is that those who exercise in outdoor natural areas and those who exercise in urban indoor areas present differences in frontal cortical thickness. In addition, this study found that those who exercised in natural outdoor areas showed better cognitive measures (as suggested by the recent studies of D'Aurizio et al., 2023 o Jiménez et al., 2022, among others). Further studies might analyze the implication of subcortical structures (hippocampus, anterior insula, among others) in these explored processes and the association between green exercise and the connectivity between the main brain networks related to attention and executive processes (Salience Attention Network, Central Executive Network, and the Default Mode Network). Furthermore, studies exploring sex or age differences in these processes might provide a wider information. Finally, there is a need for longitudinal studies to explore the development of the associations between brain, cognition, PA, and nature exposure across different ages.

CRediT authorship contribution statement

Antonio Baena-Extremera: Writing – review & editing, Project administration, Investigation, Conceptualization. **Cristina Martín-Pérez:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis. **Andrés Catena:** Writing – review & editing,

Methodology, Formal analysis. **Julio Fuentesal-García:** Writing – review & editing, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.mhpa.2024.100649.

Data availability

Data will be made available on request.

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